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Wax Deposition in Pores and the Permeability Change for High-wax Crude in a Sandstone Reservoir

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Wax deposition in pay zone is an important factor that influences well productivity for high-wax crude oilfield. Taking Hongze oilfield as an example, rheological property of the crude and effective flow permeability at different temperature was measured. Flow pattern and process of wax deposition in porous medium was studied. Experiment results show that seepage flow of high-wax crude is significantly sensitive to temperature. It is non-Darcy flow and there exist threshold pressure gradient when temperature is lower than wax appearance temperature. According to experimental data, seepage flow and deposition patterns were proposed.

Keywords: high-wax oil, permeability, porous medium, seepage flow, threshold pressure gradient

1. INTRODUCTION

High-wax crude with high pour point is a temperature-sensitive fluid. There are plenty of researches on rheological properties, wax participation, flow in the wellbore, and transportation pipe. For the temperature sensitive of the waxy crude, Ajienka and Ikoku (1995) fingered that waxy oil had different flow behavior at different temperature range. Kane' et al. (2004) mentioned that at high temperatures, crude oil exhibited Newtonian flow behavior because the paraffin was in the molten state. Below the wax appearance temperature (WAT), wax began to crystallize. Further cooling would lead to gelling below the critical pour-point.

For the rheological properties of waxy oil, Sifferman (1979) found that the pour point does not enough completely describe the flow characteristics of the waxy oil, so viscosity and gel strength should also be considered. Ajienka and Ikoku (1991) discussed the effect of temperature on the rheological properties of waxy oil in Nigerian oilfields. It is found that the generalized yield stress model is suitable for the waxy oil below the pour point. The Caisson model is adequate for waxy oil at low temperature and low shear rate. Waxy oil will behave as Newtonian fluid while the temperature is higher than the pour point. Rønningsen (1992) obtained time-dependent property of North Sea waxy oil. Slight temperature cycling ($\pm 1^\circ\text{C}$) induced a rebuilding of a weak structure, but did not restore the original gel strength completely. Li et al. (2009) studied the structural properties of gelled waxy oil from Daqing oilfield by experimental method. He found that when

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the stress acting on the waxy oil got to the yield stress, the corresponding structure state deviated far from the linear viscoelastic regime, which means that the structure of the waxy crude oil had been greatly destroyed.

For the wax deposition in porous medium, Dos Santos et al. (2004) used the coldfinger apparatus to study the deposition of Brazilian oil. Results showed that there were several elements, including temperature difference, cooling surface, nucleation kinetics and crystals growth on the coldfinger surface, could affect the rate of wax deposition. Zougari (2010) found that wax deposition was affected by a wide range of parameters including shear, temperature, pressure and time. For the flow at different temperatures, Sola (2007) discussed the temperature effect on two phase flow.

Hongze oil reservoir is a mid-porosity and low-permeability sandstone reservoir with high wax content and high pour-point crude oil. The dynamic data of the test wells indicates that oil production rate declines quickly. Because the reservoir temperature is 2–3°C higher than the pour point, temperature sensitivity may be a factor restricting oil field development efficiency during water injection. Consequently, it is vital and urgent to study the flow characteristics for this waxy crude oilfield.

2. RHEOLOGICAL PROPERTIES OF WAXY CRUDE OIL

Pour point and viscosity of the waxy crude oil are important factors to seepage flow and well productivity. Following the method of GB/T510-83 (China National Standard), the pour point of Hongze crude oil is measured to be 38.5°C. PVT Instrument (RUSKA 2730) with a capillary viscometer is used to measure the rheological character of the oil. The capillary inner diameter is 0.74 mm, and oil flow state in capillary is very similar to that in rock pores. Viscosity results are shown in Figure 1. Under steady flow state, flow rate and pressure difference across both ends of capillary are recorded during the test. Then, shear stress and shear rate is calculated by the formula (Li, 1994):

$$\text{Shear rate: } \gamma = \frac{4Q}{\pi r^3}, \quad (1)$$

$$\text{Shear stress: } \tau = \frac{\Delta Pr}{2L}. \quad (2)$$

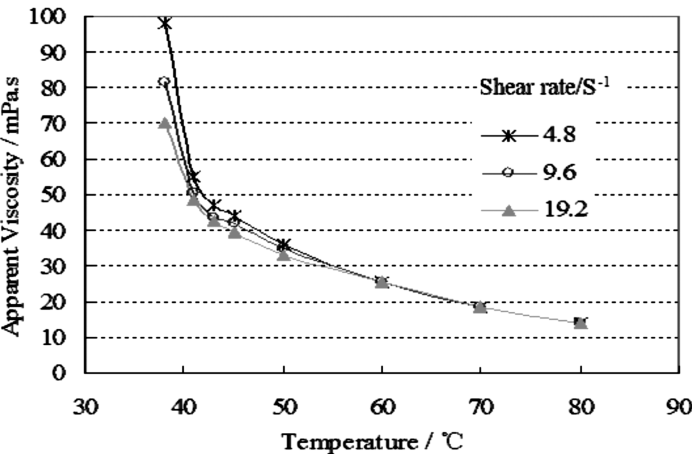


FIGURE 1 Viscosity-temperature curves of Hongze crude.

TABLE 1
Rheological Parameter of the Waxy Crude From Hongze Reservoir

Type of Fluid	Consistency Coefficient (C)	Rheological Index (n)	Temperature, °C
Newtonian fluid	27.335	1	80
	32.608	1	70
	36.794	1	60
Pseudoplastic fluid	85.418	0.8573	50
	114.15	0.8142	45
	150.72	0.7464	43
	178.47	0.7364	41
	368.38	0.6314	38

The rheological equation can be expressed in power law:

$$\tau = C\gamma^n. \quad (3)$$

Under constant temperature, by changing the flow rate, relation curves between τ and γ are tested and calculated. The slope and intercept will indicate the rheological model and rheological parameters of the waxy crude oil, as listed in Table 1.

According to data in Figure 1 and Table 1, WAT and abnormal point temperature is 60°C and 50°C, respectively. When temperature is above 60°C, the particle of waxy crystal would dissolves in the oil and the crude behaves as Newtonian fluid. Below 50°C, waxy crystal exists in the oil and crude viscosity changes with shear rate. Therefore, the crude is Newtonian fluid if above 50°C, and it is non-Newtonian fluid if below 50°C.

In the range of 38°C to 50°C, high pour-point oil behaves as pseudoplastic fluid and has certain thixotropy. As the temperature decreases, more wax crystallizes in crude causing changes in both inner physical structure and macro physical property.

3. PERMEABILITY CHANGES WITH TEMPERATURE DURING SEEPAGE FLOW

To analysis the waxy crude oil flow in porous medium at different temperature, core flooding tests were conducted at 80, 70, 60, 50, 40, and 38°C separately. Oil effective permeability and fluidity k/μ were measured to represent the flow ability of oil.

As shown in Figures 2 and 3, both oil effective permeability and fluidity k/μ reduce with temperature declines, because of wax deposition. Even if the temperature is above pour point, permeability shows a decreasing trend. For high-permeability cores, the permeability change is small, while for low-permeability cores, the change is much sharper with temperature.

Wax Deposition in Pores

The decline of oil effective permeability and flow capability indicates that wax deposition happened in rock pores. Some paraffin crystal could participate in pores when the temperature is below WAT, Even if temperature is still above the pour point.

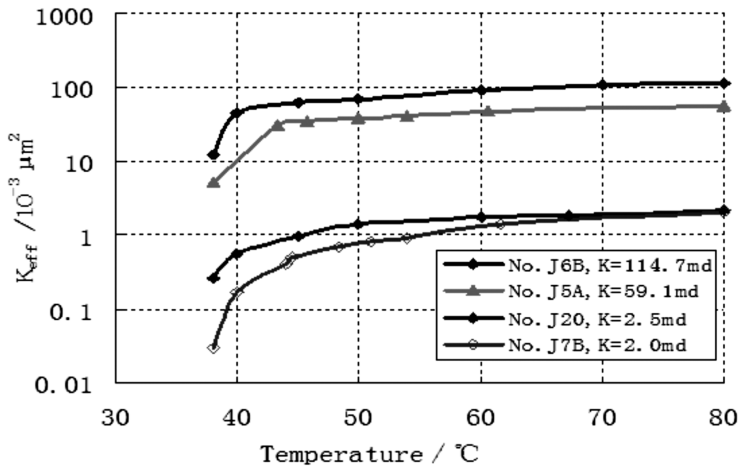


FIGURE 2 Oil effective permeability under different temperatures.

Rock pore is not only an original oil stored spaces and seepage channels, but also spaces for wax deposition. The deposition could decrease porosity and permeability. Theoretically, the amount of wax deposition relates to many factors, such as crude composition, temperature, driving pressure gradient, oil flow rate, and so on.

To study the permeability sensitivity to temperature, rate of permeability change is defined, which indicates the influence degree of temperature to permeability:

$$R_k = \frac{\Delta K}{K \Delta T}. \quad (4)$$

The experiment data show that R_k is $5.03 \times 10^{-3} \text{C}^\circ$ if temperature $> 70^\circ \text{C}$, and R_k is $70.51 \times 10^{-3} \text{C}^\circ$, if temperature is $> 38 \sim 45^\circ \text{C}$.

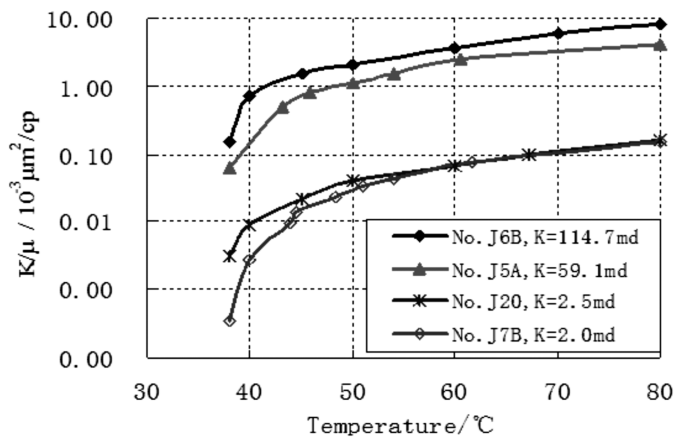


FIGURE 3 Fluidity coefficient versus temperature curve.

4. DEPOSITION AND ACCUMULATION EFFECT ON PERMEABILITY

Wax crystal is the heavier component in the crude oil. As more oil flow through the core, more wax crystal will deposit in the pore. The wax participates in the pores gradually and permeability falls gradually. Figure 4 is the permeability change with oil volume at temperature of 50°C. Permeability gradually falls with the increase of oil volume.

For waxy crude oil seepage flow, effective permeability is a multivariable function:

$$K_e = f(K_0, T, V_C). \quad (5)$$

Paraffin participation may happen in long time flow, leading to a decline of K_e . According to the measured data, the following empirical formulas are derived:

$$K_e = K_0 e^{-0.0178 V_C}, \quad (\text{for cores } K_g = 88 \times 10^{-3} \mu\text{m}^2), \quad (6)$$

$$K_e = K_0 e^{-0.058 V_C}, \quad (\text{for cores } K_g = 56 \times 10^{-3} \mu\text{m}^2), \quad (7)$$

$$K_e = K_0 e^{-0.1246 V_C}, \quad (\text{for cores } K_g = 10.4 \times 10^{-3} \mu\text{m}^2). \quad (8)$$

These formulas can be used for prediction of the permeability change of different rocks. Dynamic equilibrium. When driving pressure gradient is large enough to overcome the stress of the wax crystal deposition, the rate of paraffin deposition is approximately equal to the rate of wax destroy. The oil flow will reach a dynamically equilibrium state that oil permeability is almost stable.

5. THRESHOLD PRESSURE GRADIENT

At a given temperature, threshold pressure gradient (TPG) can be determined by measuring the relation between flow rate and pressure gradient. Figure 5 is test results of TPG at different temperatures. As temperature drops, the threshold pressure gradient rises rapidly.

In addition, the threshold pressure gradient of high waxy crude is 6–10 times that of light oil, which indicates that wax deposition raises the threshold pressure gradient and makes flow more difficult. Furthermore, deposition of wax crystal is more serious in low-pressure gradient conditions, resulting in high threshold pressure gradient.

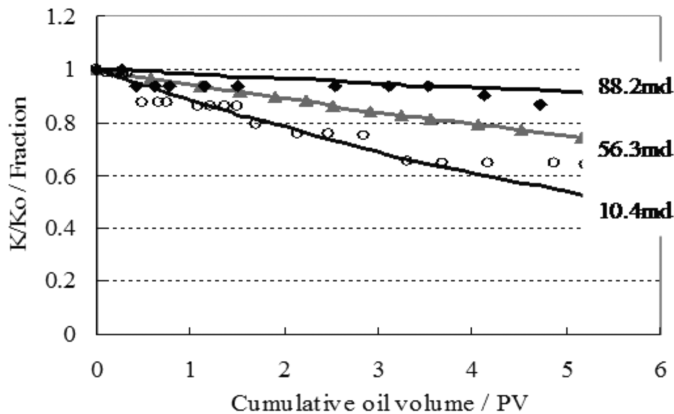


FIGURE 4 Permeability curve during long-term oil flow.

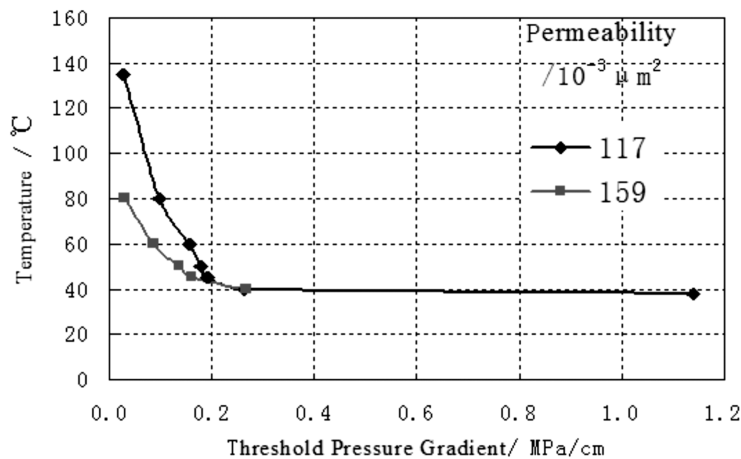


FIGURE 5 Threshold pressure change with temperature.

6. FLOW PATTERN OR MODEL UNDER DIFFERENT TEMPERATURE AND PRESSURE GRADIENT

According to above measurement and analysis, flow patterns are set for seepage flow of high-wax crude, to describe the process of paraffin participation in pores. Flow pattern under different temperature and pressure gradient is shown in Figure 6.

Zone I is Darcy flow zone, where temperature is higher than WAT and no waxy crystal appears. This is an ideal condition and the case that the crude flow does not cause the drop of oil permeability in stratum.

Zone II and zone III are zones of non-Darcy flow. Zone II is a zone with filtration and deposition. In this zone, the temperature is lower than WAT but higher than pour point, waxy crystal appears in the oil and will filtrate and deposit in the pore during the process of flowing.

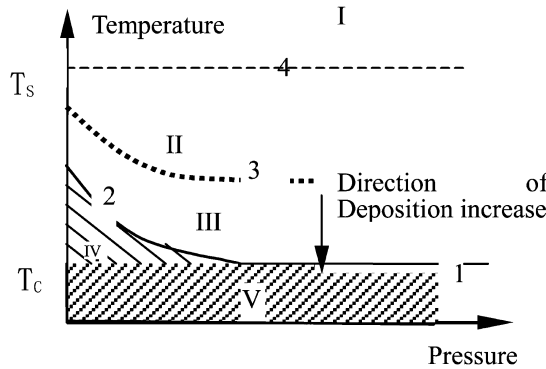


FIGURE 6 Flow pattern under different temperature and pressure gradient. Zones are the following: I = Darcy flow, II = wax deposit and filtration flow, III = adhesion deposit and filtration flow, IV = nonflowing under low pressure gradient, V = wax solid block. Lines are the following: 1 = temperature line of pour point, 2 = line of threshold pressure gradient, 3 = boundary between zone II and zone III, 4 = the lowest temperature line of waxy crystal (boundary between zone I and zone II). T_s = temperature of cloud point, T_c = temperature of pour point.

In zone III the shape of the waxy crystal is enlarged and the volume is increased with the drop of temperature. Wax deposition is attached to the surface of pores, and the wax content is the combination of filtration and adhesion sedimentation. Waxy crystal deposition leads to the drop of oil effective permeability.

Zone V is the blocked zone. At or below pour point temperature, the crude will solidify and not flow in the pores. Zone IV is a nonflowing zone under low-pressure gradient. Although temperature is higher than the pour point, oil does not flow in this range because the drive pressure gradient is less than threshold pressure gradient.

The previous flow pattern and model show that the waxy oil flow is temperature sensitive. During the process of reservoir development (i.e., water injection), the decline in temperature of pay zone will cause wax deposition, which is called cooling damage. Attention should be paid more attention to prevent this cooling damage.

In this reservoir, because temperature is lower than WAT, even though it is higher than pour point, crude flow will also cause the filtration and adhesion, leading to the drop of permeability and productivity. Consequently, to prevent wax crystal deposition in pay zone, steam injection or warm water injection is a suitable method for high-wax crude.

7. CONCLUSIONS

1. Wax deposition is an important factor leading to oil effective permeability drop and well productivity drops.
2. As soon as temperature below WAT, effective permeability of oil decreases with temperature. Some people think if the reservoir temperature is above the pour point, it is ok. This is not correct.
3. When temperature is between pour point and WAT, waxy crystal appears in waxy crude, and then filtration or adhesion happens in the pore, eventually, the effective oil permeability falls gradually.
4. TPG exists in the flow process of high waxy crude in pore medium. It is a function of temperature. The value can reach 1 MPa/m, which is 6–10 times to that in light oil reservoir.
5. Steam or warm water injection is a great method to keep high temperature and to prevent wax deposition.

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NOMENCLATURE

γ	shear rate, S^{-1}
τ	shear stress, Pa
r	capillary tube radius, m
L	capillary tube length, m
n	rheological index
Q	volume flow rate of crude oil, m^3/sec
ΔP	pressure difference across two sides of capillary tube, Pa
C	viscosity coefficient
R_k	permeability changing rate, $1/^{\circ}C$
ΔK	permeability difference, $10^{-3} \mu m^2$
ΔT	temperature difference, $^{\circ}C$
K_e	effective permeability/ $10^{-3} \mu m^2$;
K_0	original permeability, $10^{-3} \mu m^2$
V_c	cumulative oil flow, PV
T	temperature, $^{\circ}C$